Tashi Wangchuk

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Education

University of New Brunswick, Fredericton, NB, Canada **2021 - 2025**
Bachelor of Science in Math-Physics, First Class Honours in Physics Graduation: May 2025

Experience

Researcher

Druk Gyalpo Institute **Janaury 2026–Present**

- Conduct research on assessment philosophy, assessment validation, and curriculum development, while collaborating with educators to support the Bhutan Baccalaureate's evolving learning and assessment framework.

AI–STEM Data Training Specialist

Cohere Inc. **February 2025–Present**

- Labeled, ranked, audited, and corrected machine learning data with a strong focus on scientific, mathematical, and logical content.
- Completed reading and quantitative reasoning tasks with efficiency and accuracy.
- Evaluated response quality based on detailed style and content guidelines, contributing to model preference alignment.
- Identified and recommended opportunities for workflow and output optimization.
- Collaborated with cross-functional teams by providing actionable feedback on model performance and data quality.
- Maintained high attention to detail while performing repetitive and precision-driven tasks.

Associate Intern

Terrama Pte Ltd **October 2025–January 2026**

- Supported the Zhemgang Integrated Coffee Agroforestry Carbon Project under the Global Carbon Council (GCC), contributing to project registration through preparation of the Project Concept Form (PCF).
- Conducted research on carbon markets and carbon-credit quality, authored a research article (*Why Carbon-Credit Quality Matters*) for the company website, and delivered a presentation on the topic.
- Assisted in developing course outlines for the Singapore Institute of Management (SIM).

Research Assistant

UNB MRI Research Centre with Dr. Bruce J. Balcom **May 2024–May 2025**

- Developed a simple, non-invasive, non-contact method to measure electrical conductivity in water-based samples using radiofrequency (RF) probe loading.
- Combined theoretical analysis, simulations, and experimental validation to establish and refine this novel technique for conductive samples.
- Published findings in a peer-reviewed research paper and received the Science Atlantic Undergraduate Research Award for outstanding research presentation.
- Demonstrated that RF probe loading can complement or surpass conventional conductivity measurement methods, offering a robust and non-invasive alternative for applications in material characterization and biological systems.

UNB Department of Mathematics and Statistics with Dr. Edward Wilson-Ewing **May–August 2023**

- Numerically solved first-order coupled partial differential equations in effective Loop Quantum Cosmology with spherical symmetry, using Python to develop and implement algorithms. This research contributed to understanding the dynamics and behavior of complex cosmological models.
- Conducted statistical analysis on the effects of global warming on temperature distributions in Canada, showcasing data analysis and interpretation skills using Microsoft Office tools.

Tutoring Experience

UNB Department of Mathematics and Statistics **September 2023–April 2025**

- Delivered thorough feedback on coursework using Crowdmark platform for MATH1843 (Mathematics for Management).
- Tutored students one-on-one and in groups on various first and second year math courses at Math Learning Center.
- Organized and led review sessions, offering additional practice, worked examples, and comprehensive review of class material before term-tests and final exams.

Administrative Experience

UNB MRI Research Centre **September 2024–April 2025**

- Provide a broad administrative and research support to the UNB MRI Research Centre.
- Supported lab operations (chemical inventory, equipment maintenance), assisted with literature searches and reference database management, and facilitated lab meetings.
- Demonstrated independent decision-making, attention to detail, critical thinking, and proficiency in Microsoft 365, playing an integral role in the MRI Centre's success. **Supervisor**: Bruce J. Balcom.

Publications

- **Wangchuk, T.**, Aguilera, A.R., and Balcom, B.J.
  Non-Invasive, Non-Contact Conductivity Measurement Using Radiofrequency Probe Loading,
  Measurement Science and Technology (2025) 36, 105107 (12 pages).

Presentations

- Atlantic Undergraduate Physics and Astronomy Conference, St. John's, Canada, **Conductivity Measurement Using Radiofrequency Probe Loading**, January 31–February 2, 2025.
- Canadian Undergraduate Physics Conference, Vancouver, Canada, **Inductive Losses in Electrically Conducting Solutions**, October 24-27, 2024.
- Insights for Undergrads, Fredericton, Canada, **Navigating Summer Research: Key Insights for Undergraduates**, September 27, 2024.

Honours and Awards

- **Science Atlantic Undergraduate Research Award**, Presented to the student giving the best research presentation(s) at an annual Science Atlantic conference, *Science Atlantic,* January 31–February 2, 2025.
- **Dr. Margot R. Roach Scholarship**, Awarded based on Scholastic Achievements upon recommendation by the Department of Physics, *University of New Brunswick,* September 2023 - May 2025.
- **Department of Mathematics Prizes in Enriched Calculus**, Awarded to the top students who, in the opinion of the Department of Mathematics and Statistics, show the greatest promise in Mathematics, *University of New Brunswick,* September 2021 - May 2022.

- **Dean's List Award**, Awarded to the students achieving a grade-point average of 3.7 or higher and acknowledges excellent performance in what can be a very demanding degree and course load, *University of New Brunswick,* 2021 - 2022, 2023 - 2024, 2024 - 2025.
- **Royal Government of Bhutan's Scholarship**, Granted to high-achieving individuals among 13,560 grade-12 students to undertake full time undergraduate study in different parts of the world, focusing on addressing the country's priority human resource and development needs, *Ministry of Education and Skills Development, Bhutan,* September 2021 - May 2025.
- **J.D. Irving Student Scholarship**, Offered to the post-secondary student tree planters as a validation for their hard work and dedication throughout the tree-planting season apart from successful completion of criteria identified in the policy, *J.D. Irving,* 2022 - 2023.

Professional Development Certifications

- **Google AI Essentials** — Google (Coursera), 2024. Demonstrated ability to integrate AI tools into daily workflows to enhance efficiency, automate routine tasks, and increase overall productivity.
- **Google Data Analytics Professional Certificate** — Google (Coursera), In Progress. Comprehensive training in R programming, SQL, Python, and Tableau, with focus on applying AI in data analytics. Completed the following certificates under this program:
  1. **Foundations: Data, Data, Everywhere** — Gained foundational understanding of data analytics, including data ecosystems, analytical thinking, and the roles of spreadsheets, query languages, and visualization tools in data analysis.
  2. **Ask Questions to Make Data-Driven Decisions** — Learned structured thinking and problem-solving approaches for analysis scenarios, emphasizing data-driven decision-making and spreadsheet-based data organization.
- **Google Prompting Essentials Specialization** — Google (Coursera). Developed advanced prompting skills to use AI effectively across professional, analytical, and creative tasks. Completed the following certificates under this program:
  1. **Start Writing Prompts like a Pro** — Learned foundational prompting strategies, including clarity, structure, and iterative refinement to improve AI responses.
  2. **Design Prompts for Everyday Work Tasks** — Gained skills to automate routine workflows using AI, enhancing productivity in writing, summarizing, planning, and task delegation.
  3. **Speed Up Data Analysis and Presentation Building** — Applied prompting techniques to accelerate data exploration, insight generation, and slide-building using AI tools.
  4. **Use AI as a Creative or Expert Partner** — Learned to collaborate with AI for ideation, decision-support, and expert-level reasoning across technical and creative domains.

Community Involvement

Physics and Astronomy Club, **Vice–President of Events** — September 2024–May 2025
Canadian Association of Physicists (CAP) congress, **Student Volunteer** — June 2023
Parkinson Canada SuperWalk, **Volunteer** — September 2023